

Algebra I Systems Cheatsheet

Module 6

Meaning of a Solution

A solution to a system is an ordered pair that makes every equation true.

$$\begin{cases} y = 2x + 1 \\ y = -x + 7 \end{cases}$$

The graphing solution is the intersection point.

Graphing Method

- Put both equations in graphable form.
- Graph both lines.
- The intersection is the solution.

Substitution Method

Use when one equation is already solved for a variable.

$$y = 3x - 4, \quad 2x + y = 11$$

Substitute:

$$2x + (3x - 4) = 11$$

Elimination Method

Use when adding equations can eliminate a variable.

$$2x + 3y = 11$$

$$4x - 3y = 7$$

Add:

$$6x = 18$$

Number of Solutions

one solution	different slopes	intersecting lines
no solution	same slope, different intercepts	parallel lines
infinitely many	same line	identical equations

Systems Word Problems

Define variables first.

x = first quantity, y = second quantity

Look for:

total equation, rate/value equation

Quick Checks

- The answer to a two-variable system is usually an ordered pair.
- Substitute the ordered pair into both equations to check.
- Elimination works best when coefficients are already opposites or can be made opposites.
- Parallel lines have no intersection.